PART A — QUESTIONS

AF20-001
Question:
Solve: $22x + 14 = 146$
A) 5
B) 6
C) 7
D) 8

AF20-002
Question:
If $20n - 11 = 149$, what is n?
A) 6
B) 7
C) 8
D) 9

AF20-003
Question:
Simplify: $7(4x - 5) - 9x$
A) $19x - 35$
B) $28x - 35$
C) $19x - 5$
D) $37x - 35$

AF20-004
Question:
A company charges $42 per repair plus a $35 service fee. Which expression represents the total charge for r repairs?
A) $42r + 35$
B) $35r + 42$
C) $77r$
D) $42 - 35r$

AF20-005
Question:
If $f(x) = 17x - 49$, what is $f(8)$?
A) 77
B) 87
C) 136
D) 185

AF20-006
Question:
Which ordered pair satisfies the equation $y = 15x - 7$?
A) (1, 9)
B) (2, 23)
C) (3, 38)
D) (4, 54)

AF20-007
Question:
Solve: 24(x - 6) = 168
A) 11
B) 12
C) 13
D) 14

AF20-008
Question:
A pattern begins: 7, 26, 45, 64, ... What is the next number?
A) 81
B) 82
C) 83
D) 84

AF20-009
Question:
Solve: x/40 = 3
A) 43
B) 80
C) 120
D) 160

AF20-010
Question:
A line has a slope of -17 and crosses the y-axis at 14. Which equation represents the line?
A) y = 14x - 17
B) y = -17x + 14
C) y = 17x + 14
D) y = -14x + 17

AF20-011
Question:
If 9 crates contain 306 parts, how many parts are in 6 crates at the same rate?
A) 192
B) 198
C) 204
D) 216

AF20-012
Question:
Solve: 280 - 28x = 168
A) 3
B) 4
C) 5
D) 6

AF20-013
Question:
Which expression is equivalent to 96a - 44a - 18?

A) 52a - 18
B) 140a - 18
C) 52a + 18
D) 96a - 62

AF20-014
Question:
A storage rack starts with 1,600 parts. Each shipment removes 80 parts. Which equation gives the number of parts P left after s shipments?
A) P = 1,600s - 80
B) P = 80s + 1,600
C) P = 1,600 - 80s
D) P = 80s - 1,600

AF20-015
Question:
If y = -22x + 300, what is y when x = 12?
A) 26
B) 36
C) 264
D) 564

AF20-016
Question:
Solve: 34x - 60 = 22x + 84
A) 10
B) 11
C) 12
D) 13

AF20-017
Question:
Which value of x makes 18x + 4 less than 220?
A) 11
B) 12
C) 13
D) 14

AF20-018
Question:
A table shows a linear relationship.

x: 0, 1, 2, 3
y: 8, 28, 48, 68

What is the rule?
A) y = 8x + 20
B) y = 20x + 8
C) y = 28x - 20
D) y = x + 8

AF20-019
Question:
Solve: 18(x + 4) = 15x + 108
A) 10
B) 11
C) 12
D) 13

AF20-020
Question:
If a = -14 and b = 12, what is 5a + 6b?
A) -142
B) -2
C) 2
D) 142

AF20-021
Question:
Which point is located in Quadrant II?
A) (16, 9)
B) (-16, 9)
C) (-16, -9)
D) (16, -9)

AF20-022
Question:
A worker checks 240 items in 16 minutes. At the same rate, how many items can the worker check in 7 minutes?
A) 90
B) 100
C) 105
D) 120

AF20-023
Question:
Solve for t: D = 32t - 64
A) t = D + 64
B) t = (D + 64)/32
C) t = 32(D + 64)
D) t = D/32 - 64

AF20-024
Question:
What is the slope of a line that goes through (8, 15) and (14, 93)?
A) 11
B) 12
C) 13
D) 78

AF20-025
Question:

Simplify: $76x - 27 - 65x + 42$
A) $11x + 15$
B) $141x + 15$
C) $11x - 69$
D) $141x - 69$

AF20-026
Question:
If $g(x) = x^2 + 7x$, what is $g(12)$?
A) 144
B) 168
C) 228
D) 288

AF20-027
Question:
A sequence follows the rule “multiply by 5, then subtract 17.” If the first term is 10, what is the second term?
A) 33
B) 40
C) 50
D) 67

AF20-028
Question:
Solve: $24(x - 5) + 20 = 212$
A) 11
B) 12
C) 13
D) 14

AF20-029
Question:
A line crosses the y-axis at -21 and rises 10 units for every 1 unit to the right. Which equation represents the line?
A) $y = -21x + 10$
B) $y = 10x - 21$
C) $y = -10x - 21$
D) $y = 21x + 10$

AF20-030
Question:
If $28/42 = x/126$, what is x?
A) 72
B) 78
C) 84
D) 90

AF20-031
Question:
Solve: $-36x + 144 = -288$

A) -12
B) -8
C) 8
D) 12

AF20-032
Question:
A warehouse starts with 5,000 washers. Each kit uses 250 washers. Which expression gives the number of washers left after k kits?
A) 5,000 + 250k
B) 5,000 - 250k
C) 250k - 5,000
D) 5,000 ÷ 250k

AF20-033
Question:
Which equation has x = 26 as its solution?
A) 2x + 15 = 65
B) 5x - 32 = 98
C) x/13 + 8 = 11
D) 3x - 20 = 55

PART B — ANSWER KEY + EXPLANATIONS

AF20-001 — Correct: B — Explanation:
Solve 22x + 14 = 146. Subtract 14 from both sides: 22x = 132. Divide by 22: x = 6. The most common mistake is subtracting 14 correctly but forgetting to divide by 22.

AF20-002 — Correct: C — Explanation:
Start with 20n - 11 = 149. Add 11 to both sides: 20n = 160. Divide by 20: n = 8. A common mistake is subtracting 11 instead of adding it.

AF20-003 — Correct: A — Explanation:
Distribute first: 7(4x - 5) = 28x - 35. Then subtract 9x: 28x - 9x - 35 = 19x - 35. A common mistake is forgetting to multiply -5 by 7.

AF20-004 — Correct: A — Explanation:
The company charges $42 for each repair, so that part is 42r. The $35 service fee is added one time. The total charge is 42r + 35. A common mistake is treating the service fee as the per-repair charge.

AF20-005 — Correct: B — Explanation:
Substitute 8 for x: f(8) = 17(8) - 49 = 136 - 49 = 87. A common mistake is adding 49 instead of subtracting it.

AF20-006 — Correct: C — Explanation:
Test the choices in y = 15x - 7. For (3, 38), 15(3) - 7 = 45 - 7 = 38, which matches the y-value. A common mistake is choosing a point that is close but does not make the equation true.

AF20-007 — Correct: C — Explanation:
Solve 24(x - 6) = 168. Divide both sides by 24: x - 6 = 7. Add 6: x = 13. A common mistake is adding 6 before undoing the multiplication by 24.

AF20-008 — Correct: C — Explanation:
The pattern increases by 19 each time: 7, 26, 45, 64. Add 19 to 64: 83. A common mistake is assuming the pattern increases by 18 because the numbers are close together.

AF20-009 — Correct: C — Explanation:
$x/40 = 3$ means x divided by 40 equals 3. Multiply both sides by 40: $x = 120$. A common mistake is adding 40 and 3 instead of multiplying.

AF20-010 — Correct: B — Explanation:
Slope-intercept form is $y = mx + b$. The slope is -17 and the y-intercept is 14, so the equation is $y = -17x + 14$. A common mistake is switching the slope and y-intercept.

AF20-011 — Correct: C — Explanation:
Find the unit rate: 306 parts ÷ 9 crates = 34 parts per crate. For 6 crates: $34 \times 6 = 204$. A common mistake is multiplying 306 by 6 without finding the rate first.

AF20-012 — Correct: B — Explanation:
Solve $280 - 28x = 168$. Subtract 280 from both sides: $-28x = -112$. Divide by -28: $x = 4$. A common mistake is losing the negative sign after subtracting 280.

AF20-013 — Correct: A — Explanation:
Combine like terms: $96a - 44a = 52a$. The constant -18 stays the same. The expression becomes $52a - 18$. A common mistake is combining the constant with the variable terms.

AF20-014 — Correct: C — Explanation:
The rack starts with 1,600 parts. Each shipment removes 80 parts, so s shipments remove 80s parts. The number left is $P = 1{,}600 - 80s$. A common mistake is adding 80s even though parts are being removed.

AF20-015 — Correct: B — Explanation:
Substitute $x = 12$ into $y = -22x + 300$: $y = -22(12) + 300 = -264 + 300 = 36$. A common mistake is treating $-22(12)$ as positive 264.

AF20-016 — Correct: C — Explanation:
Solve $34x - 60 = 22x + 84$. Subtract 22x: $12x - 60 = 84$. Add 60: $12x = 144$. Divide by 12: $x = 12$. A common mistake is adding 60 to only one side.

AF20-017 — Correct: A — Explanation:
Solve $18x + 4 < 220$. Subtract 4: $18x < 216$. Divide by 18: $x < 12$. The only listed value less than 12 is 11. A common mistake is choosing 12, but $18(12) + 4 = 220$, not less than 220.

AF20-018 — Correct: B — Explanation:
The y-values increase by 20 each time x increases by 1, so the slope is 20. When $x = 0$, $y = 8$, so the y-intercept is 8. The rule is $y = 20x + 8$. A common mistake is using 8 as the slope because it is the first y-value.

AF20-019 — Correct: C — Explanation:
Distribute: $18x + 72 = 15x + 108$. Subtract 15x: $3x + 72 = 108$. Subtract 72: $3x = 36$. Divide by 3: $x = 12$. A common mistake is forgetting to multiply 4 by 18.

AF20-020 — Correct: C — Explanation:
Substitute $a = -14$ and $b = 12$: $5a + 6b = 5(-14) + 6(12) = -70 + 72 = 2$. A common mistake is treating $5(-14)$ as positive 70.

AF20-021 — Correct: B — Explanation:
Quadrant II has a negative x-coordinate and a positive y-coordinate. The point (-16, 9) fits this description. A common mistake is choosing (16, -9), which is Quadrant IV.

AF20-022 — Correct: C — Explanation:
240 items in 16 minutes means $240 \div 16 = 15$ items per minute. In 7 minutes: $15 \times 7 = 105$. A common mistake is multiplying 240 by 7 without finding the rate first.

AF20-023 — Correct: B — Explanation:
Start with $D = 32t - 64$. Add 64 to both sides: $D + 64 = 32t$. Divide by 32: $t = (D + 64)/32$. A common mistake is dividing D by 32 first and then adding 64.

AF20-024 — Correct: C — Explanation:
Slope equals change in y divided by change in x. From (8, 15) to (14, 93), y increases by 78 and x increases by 6. The slope is $78 \div 6 = 13$. A common mistake is using 78 as the slope without dividing by the change in x.

AF20-025 — Correct: A — Explanation:
Combine like terms: $76x - 65x = 11x$, and $-27 + 42 = 15$. The simplified expression is $11x + 15$. A common mistake is writing $11x - 15$ because the positive 42 is missed.

AF20-026 — Correct: C — Explanation:
Substitute 12 for x: $g(12) = 12^2 + 7(12) = 144 + 84 = 228$. A common mistake is calculating 12^2 as 24 instead of 144.

AF20-027 — Correct: A — Explanation:
Start with 10. Multiply by 5: 50. Then subtract 17: 33. A common mistake is subtracting 17 first and then multiplying.

AF20-028 — Correct: C — Explanation:
Solve $24(x - 5) + 20 = 212$. Distribute: $24x - 120 + 20 = 212$. Combine: $24x - 100 = 212$. Add 100: $24x = 312$. Divide by 24: $x = 13$. A common mistake is forgetting to combine -120 and +20.

AF20-029 — Correct: B — Explanation:
The line rises 10 units for every 1 unit to the right, so the slope is 10. It crosses the y-axis at -21, so the equation is $y = 10x - 21$. A common mistake is switching the slope and intercept.

AF20-030 — Correct: C — Explanation:
Use the proportion $28/42 = x/126$. Since $42 \times 3 = 126$, multiply 28 by 3: $x = 84$. A common mistake is adding 84 to the numerator because 42 becomes 126.

AF20-031 — Correct: D — Explanation:
Solve $-36x + 144 = -288$. Subtract 144 from both sides: $-36x = -432$. Divide by -36: $x = 12$. A common mistake is giving -12 because both sides contain negative numbers.

AF20-032 — Correct: B — Explanation:
The warehouse starts with 5,000 washers. Each kit uses 250 washers, so k kits use 250k washers.

The number left is 5,000 - 250k. A common mistake is adding 250k even though washers are being used.

AF20-033 — Correct: B — Explanation:
Test x = 26. In choice B, 5(26) - 32 = 130 - 32 = 98, so the equation is true. A common mistake is choosing a nearby equation without checking both sides exactly.